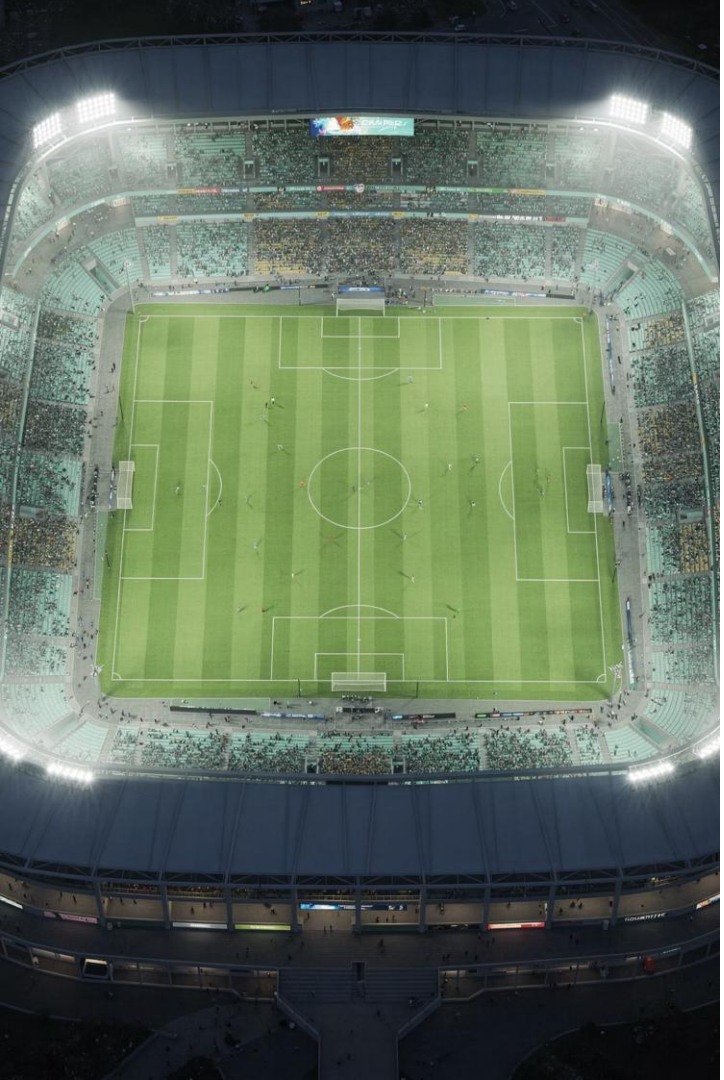


Multidimensional Unsupervised Learning

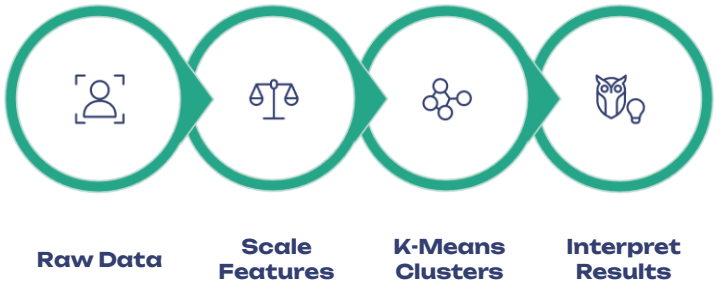
Dr. Marcelo Feighelstein

RECAP

Quick Recap: Clustering Without Labels

In unsupervised learning, we work only with features **X** — no target label **y**. The algorithm searches for hidden structure on its own. K-means assigns each sample to one of K clusters, and the resulting labels (0, 1, 2...) are arbitrary numbers with no inherent meaning.

A clustering result is not the final answer — it is the **beginning** of the analysis.



From Iris Flowers to FIFA Players

🌸 Iris Dataset

Small and clean — just 4 features:

- Sepal length
- Sepal width
- Petal length
- Petal width

Easy to visualize and reason about. A great teaching tool, but not a real-world challenge.

⚽ FIFA Dataset

Rich and realistic — dozens of features:

- Age, Overall, Potential, Wage, Value
- Finishing, Passing, Dribbling, Tackling
- GKDiving, GKHandling, GKReflexes
- Strength, Stamina, Sprint Speed

Much closer to what data scientists face in practice.

Key question: Can clustering discover player types without ever looking at the Position column?



Why Cluster Football Players?

Football players have very different profiles depending on their role. Clustering lets us ask: **can the data discover these groups automatically?**

Defensive Roles

Goalkeepers, center-backs, full-backs — strong in tackling and positioning.

Physical Profiles

Fast wingers, powerful strikers, tireless box-to-box midfielders.

Elite vs Developing

Expensive veterans vs. young high-potential prospects with low wages.

Important Warning

The algorithm may find groups that differ from what we expected as humans.

FIFA Dataset: What Features Do We Have?

Each FIFA19 player is described by dozens of numerical columns. Here is a selection of the most relevant groups:

Basic Info

Age, Nationality, Club, Position

Technical Skills

BallControl, Dribbling, Finishing, ShortPassing, Crossing

Ability

Overall, Potential

Physical Skills

Acceleration, SprintSpeed, Strength, Stamina

Market

Value (€), Wage (€/week)

GK Skills

GKDiving, GKHandling, GKKicking, GKPositioning, GKReflexes

Each player is a point in a high-dimensional numerical space — one dimension per feature.

PREPROCESSING

Preparing FIFA Data for Clustering

Raw data is rarely ready to cluster. We need to clean and prepare it carefully before running any algorithm. Here are the typical steps in order:

01

Select useful numerical features

Drop identifiers like Name, Photo URL, Club Logo.

02

Handle missing values

Drop rows or impute where data is missing.

03

Encode or drop categorical columns

Ignore Position for now — that's what we want to discover!

04

Scale with StandardScaler

All features get zero mean and unit variance.

⚠️ Why Scale?

K-means is distance-based. Features with large numeric ranges — like Wage (€300,000/week) or Value (€100M) — will **dominate** the distance calculation and distort the clusters. StandardScaler ensures every feature contributes equally.

Why High-Dimensional Data Is Hard

A football player in our dataset is not described by two numbers. They are described by potentially **40 or more features** simultaneously. This creates a high-dimensional feature space that our brains simply cannot picture.

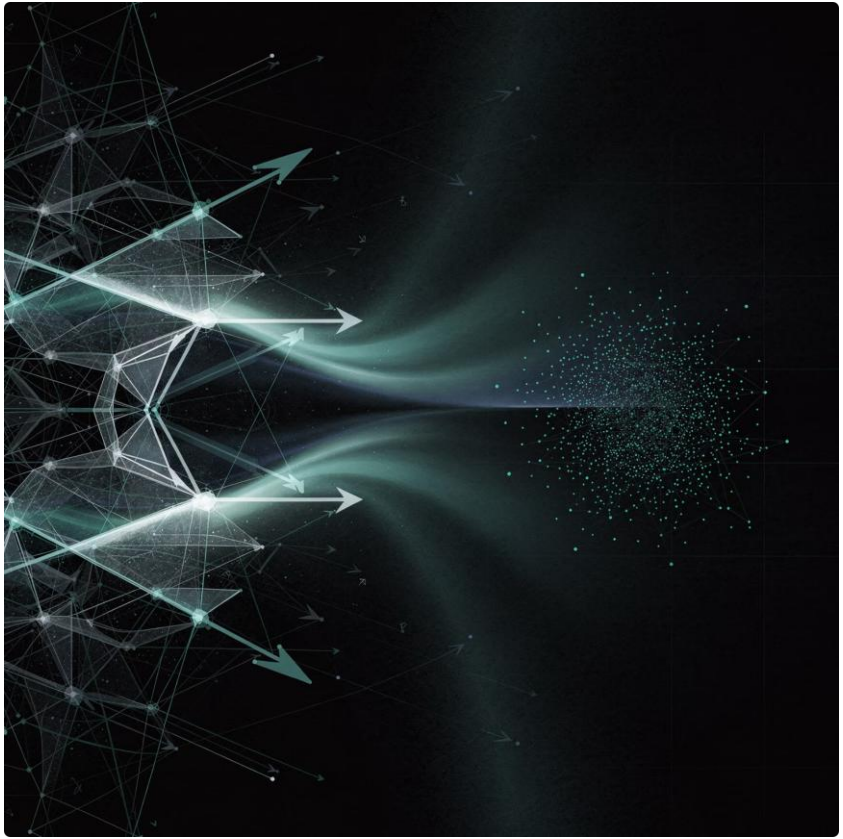
2D or 3D

Humans visualize these easily — scatter plots, 3D spinning charts.

10, 40, or 80 dimensions

Completely impossible to plot directly. We need a different approach.

Solution: **Dimensionality Reduction** — compress many features into 2 dimensions for visualization.



DIMENSIONALITY REDUCTION

Dimensionality Reduction: The Big Idea

Dimensionality reduction transforms many features into a smaller number of new dimensions — ideally just 2, so we can draw a scatter plot. Instead of plotting Dribbling vs Finishing vs GKDiving vs… we project players onto a 2D map.

Visualize Structure

See which players are similar and which are far apart — at a glance.

Detect Groups

Check whether K-means clusters are actually separated in a low-dimensional view.

Find Outliers

Spot unusual player profiles that fall far outside the main groups.

But: Information Is Lost

A 2D plot is always a simplified view. Not everything is preserved.

PCA: Principal Component Analysis

PCA finds new axes — called **principal components** — that summarize the strongest directions of variation in the data. Think of it as rotating your coordinate system so that the most "interesting" spread is visible first.

PC1

Captures the single largest source of variation across all players.

PC2

Captures the second largest direction, perpendicular to PC1.

Plot

Each player becomes a dot at position (PC1 score, PC2 score).

In FIFA: PCA often reveals a strong separation between goalkeeper-skill players and field-skill players along PC1.

How PCA Works — Step by Step

01

Start with many features

Each player is a point in a high-dimensional space (e.g. 40+ features).

02

Find the direction of most variance

PC1 is the axis along which the data spreads the most — the "most informative" direction.

03

Find the next best direction

PC2 is perpendicular to PC1 and captures the next largest spread.

04

Project everyone onto 2D

Each player gets a (PC1, PC2) coordinate and can be plotted on a 2D scatter plot.

No information about positions or labels is used — PCA is purely mathematical.

APPLIED EXAMPLE

FIFA Players in 2D Using PCA

Here is the step-by-step workflow to create a PCA visualization of FIFA players in Python:

01

Select numerical features

Exclude name, photo, club logo, position.

02

Scale with StandardScaler

All features on the same scale.

03

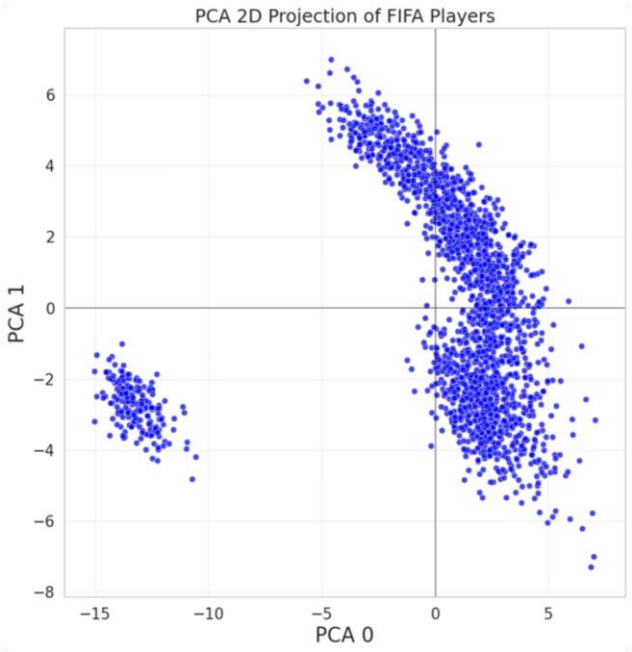
Apply PCA(n_components=2)

Compress all features to two dimensions.

04

Plot PCO vs PC1

Color each dot by cluster label.



What PCA Can and Cannot Tell Us

PCA is a powerful first step, but it has real limitations. Understanding both sides helps you use it responsibly.

PCA Strengths

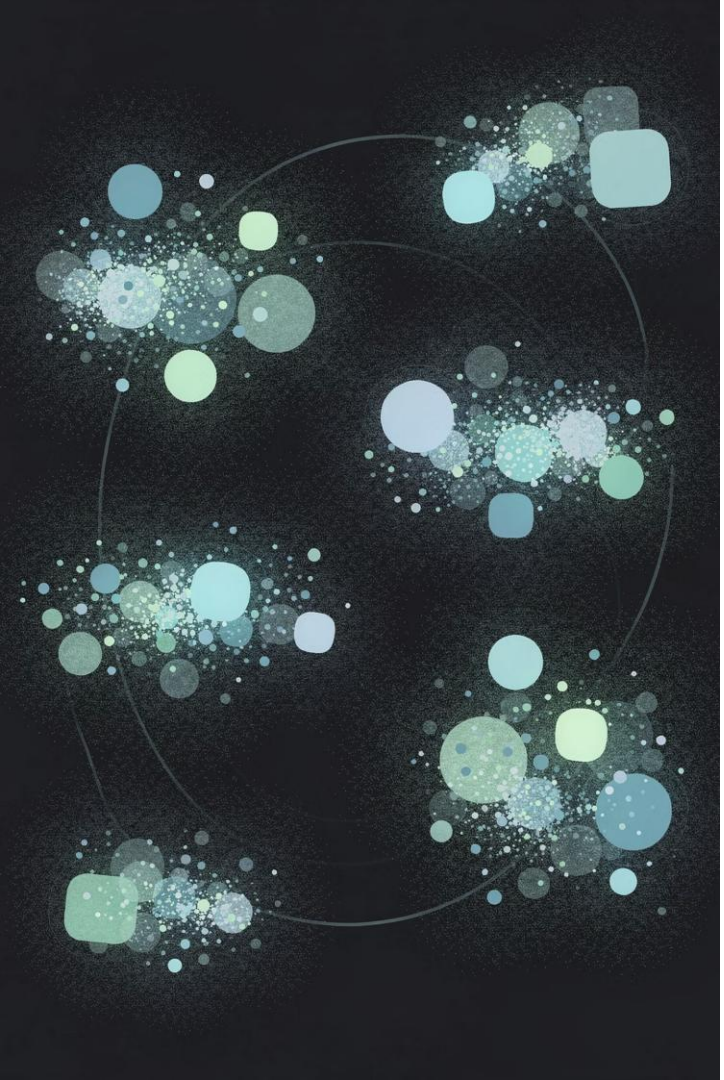
- Fast to compute, even on large datasets
- Gives a simple, interpretable 2D view
- Preserves large-scale variation well
- Components sometimes have interpretable meaning



PCA Limitations

- Linear only — misses curved or non-linear patterns
- A 2D plot always loses information from higher dimensions
- Seeing clusters in PCA is not proof the clustering is correct
- Components can be hard to interpret in complex datasets

PCA is a **visualization and exploration tool**, not a final conclusion. Always validate findings with feature-level analysis.



t-SNE: A Non-Linear 2D Map

Core Idea

If two players are very similar in the original high-dimensional space, t-SNE tries to place them close together in 2D. It focuses especially on **preserving local neighborhoods** — players who are similar stay near each other on the map.

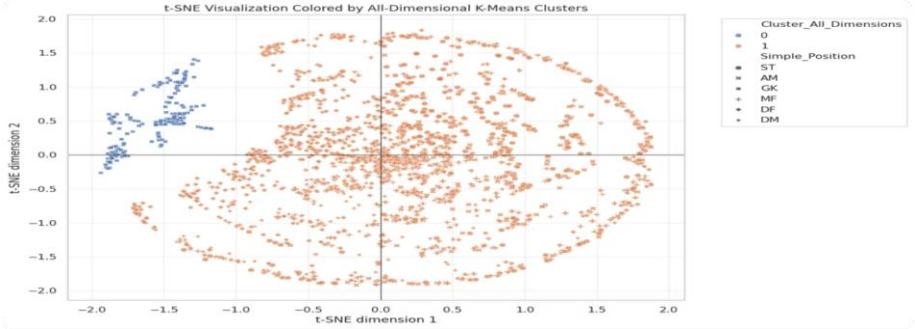
✅ Good for

- Visual exploration of groups
- Finding possible subclusters
- Spotting unusual player profiles

❌ Not good for

- Interpreting the axes
- Measuring exact distances between groups
- Comparing cluster sizes directly

t-SNE Applied to FIFA Players



What the Plot Reveals

- The **small blue island** = Goalkeepers (Cluster 0) — completely separated from all other positions
- The **large orange mass** = all field players (Cluster 1) — attackers, midfielders, defenders grouped together

The axes have no meaning — only the **relative distances** between players matter.

COMPARISON

PCA vs t-SNE: Which One Should You Use?

Both methods reduce dimensions for visualization, but they work very differently and serve different purposes.

Aspect	PCA	t-SNE
Main idea	Linear projection onto new axes	Preserves local neighborhood similarity
Speed	Fast — scales well to large data	Slow — computationally expensive
Axes interpretable?	Partially yes but difficult	No — axes have no meaning
Best for	Global structure, first overview	Local clusters, visual exploration
Risk	Misses non-linear patterns	Can exaggerate separation visually

Rule of thumb: Use PCA for a first global view. Use t-SNE to explore local structure and possible subgroups — but never treat it as ground truth.

K-MEANS ON FIFA

Clustering FIFA Players with K-Means

Once the data is scaled, running K-means on FIFA players follows the same workflow we already know — with one critical addition: **interpreting the result**.

01

Select features

Numerical columns only, no Position or name.

02

Scale with StandardScaler

Critical — K-means is distance-based.

03

Choose K and run KMeans

Use elbow method or silhouette score to guide K selection.

04

Add cluster label to DataFrame

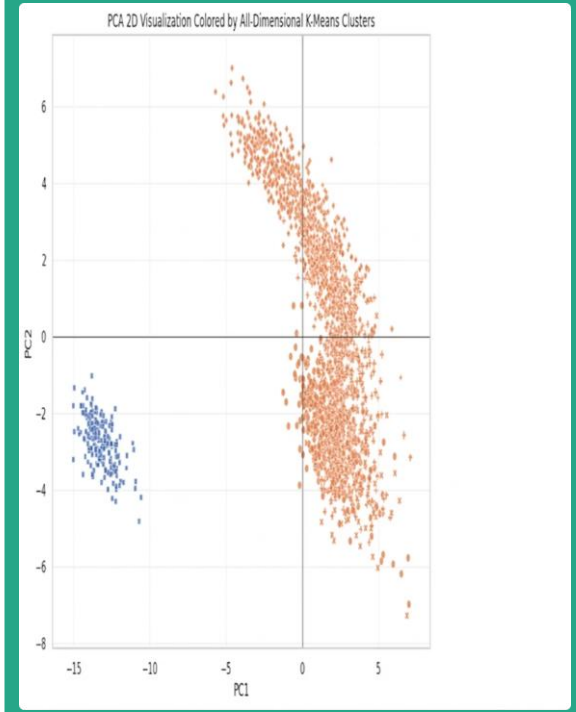
```
df["cluster"] = kmeans.fit_predict(X_scaled)
```

05

Interpret the clusters

Analyze feature averages per cluster — this is where the insight lives.

Result with K = 2



Running K-means with two clusters on the full FIFA numerical feature set produces the following result:

- **Cluster 0**
- **Cluster 1**

INTERPRETATION METHOD 1

Interpret Clusters with Centroids

A **centroid** is the average feature profile of all players in a cluster — it tells us what a typical member of that cluster looks like. By comparing centroids across clusters, we can name and understand what each group represents. In Python: `df.groupby("cluster")[features].mean()`

Cluster	index	Age	Overall	Potential	Value	Wage	International Reputation	Weak Foot	Skill Moves	Crossing	...
0	1053.293277	27.028057	77.991530	80.136051	1.300433e+07	45905.240868	1.753838	3.261514	3.114346	65.633669	...
1	950.537688	29.391960	78.633166	80.291457	1.041294e+07	36743.718593	1.768844	2.688442	1.000000	14.949749	...

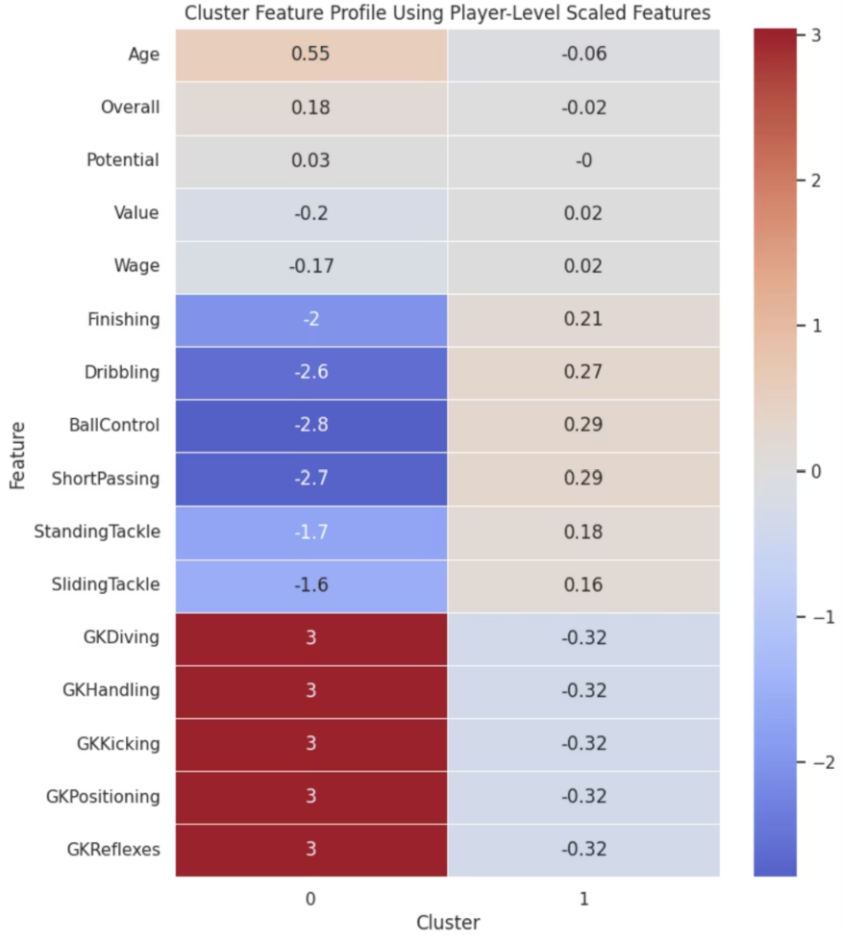
2 rows × 47 columns

Centroid table: average feature values per cluster (2 rows × 47 columns)

Compare these key features across clusters:

- **GK skills:** GKDiving, GKHandling, GKReflexes
- **Outfield skills:** Finishing, Dribbling, ShortPassing, BallControl
- **Overall profile:** Overall, Potential, Wage, Value

Cluster Feature Profile: Scaled Centroids



How to Read This Chart

Each cell shows the average standardized feature value for players in that cluster. Values near 0 mean average. Positive (red) = above average. Negative (blue) = below average.

🟥 Cluster 0 — Goalkeepers

GKDiving, GKHandling, GKKicking, GKPositioning, GKReflexes all score **+3**. All outfield skills (Finishing, Dribbling, BallControl) score around **−2 to −3**.

🟦 Cluster 1 — All Other Players

GK skills score **−0.32** (far below average). Outfield skills like Finishing (+0.21), Dribbling (+0.27), BallControl (+0.29) are **above average**.

⚖️ General Stats

Age, Overall, Potential, Wage, Value — **nearly identical** between clusters. The split is purely about skill type, not player quality.



The Main Result

FIFA Clustering: Two Groups Emerge

Cluster 0 — Goalkeepers

High GK Diving, GK Handling, GK Reflexes, GK Positioning, GK Kicking. Low finishing, dribbling, crossing, ball control. A numerically **unique** profile unlike any other position.

Cluster 1 — All Other Players

Attackers, midfielders, and defenders all land here. They share enough field-player attributes to form one large group when K = 2. The algorithm sees their similarities more than their differences.

Key insight: Clustering finds the *strongest numerical structure* in the selected feature space — not necessarily the human categories we expected. Goalkeepers are simply more different from everyone else than attackers are from defenders.

METHOD 2 + WHAT'S NEXT

Explain Clusters with Classification & Refine the Analysis

Method 2: Train a Classifier on Cluster Labels

After clustering, treat the cluster label as a temporary target variable and train a classifier. Its feature importance ranking tells us *which features drove the separation*.

01

Run K-Means, save labels

02

Train Decision Tree or Random Forest on labels

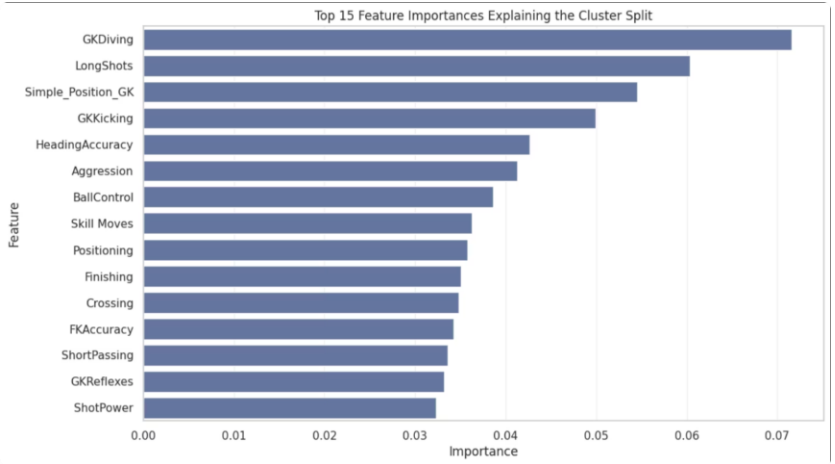
03

Read feature importances

04

Top features explain the cluster split

⚠ This does NOT mean the cluster labels are true labels. It is only a tool for *explaining* the result.



Top 15 feature importances from a classifier trained on cluster labels

GKDiving dominates — confirming the split is almost entirely driven by goalkeeper-specific skills. LongShots and Simple_Position_GK follow, reinforcing the same story.



Thank You!

Go explore data — get your hands dirty, break things, and learn by doing.

Email

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Office Hours

Per Request

Location

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